

MDAP Intern Presentation:

Music Score Analysis through Natural Language Interfaces

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Credit: Mohamed_hassan@pixabay

 Claude

 OpenAI

 Gemini

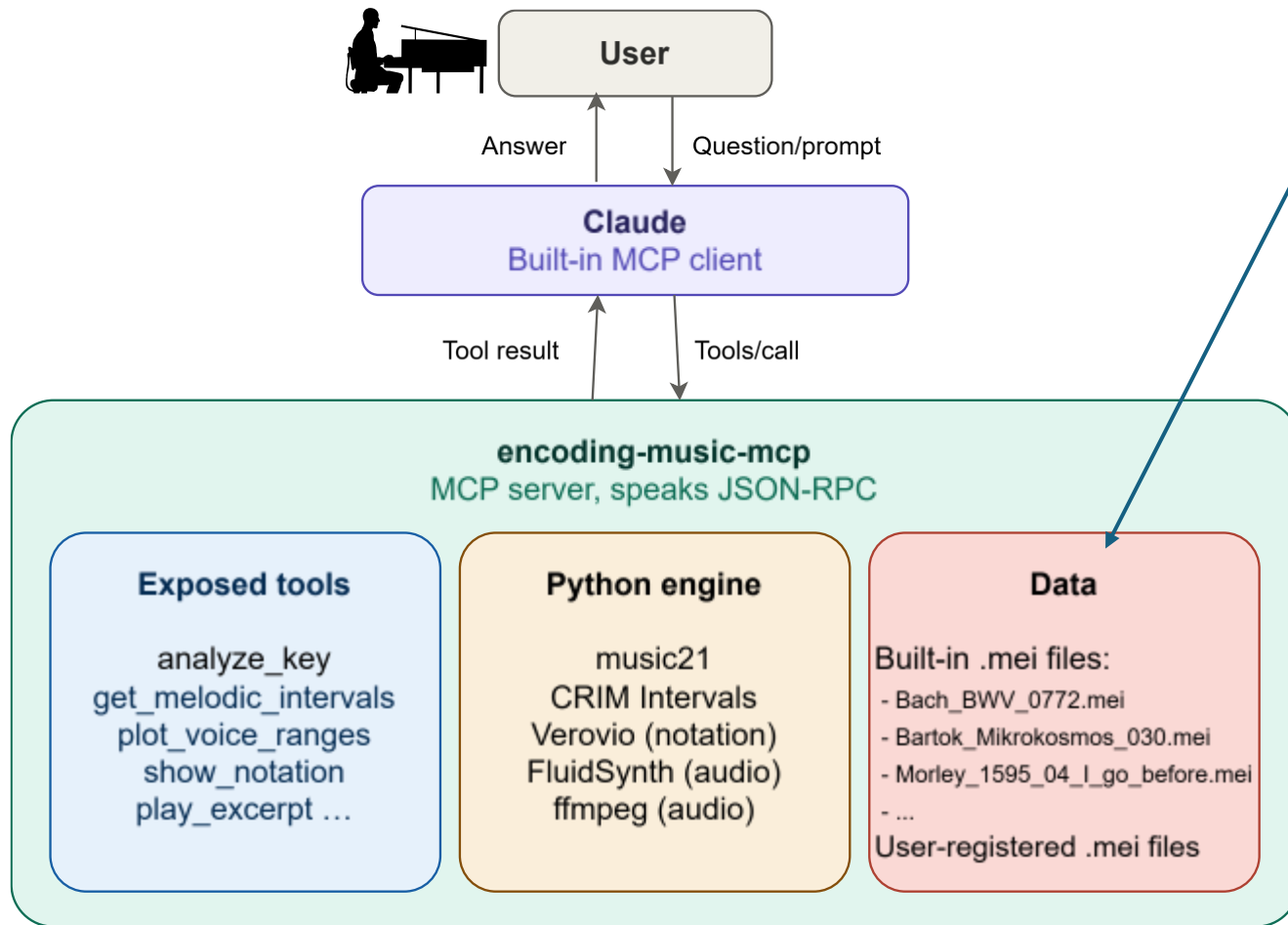
 deepseek

 Grok



Credit: Mohamed_hassan@pixabay



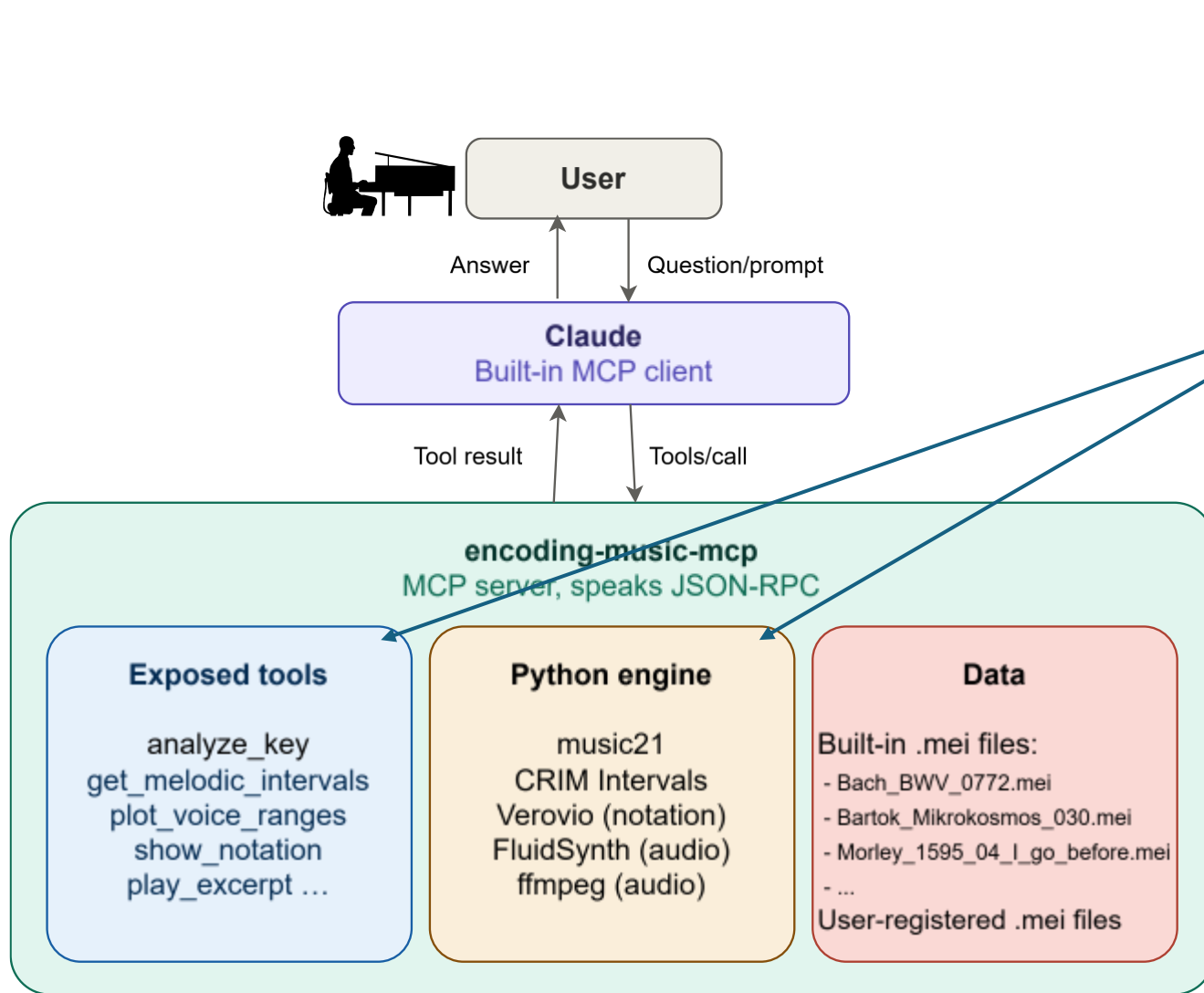


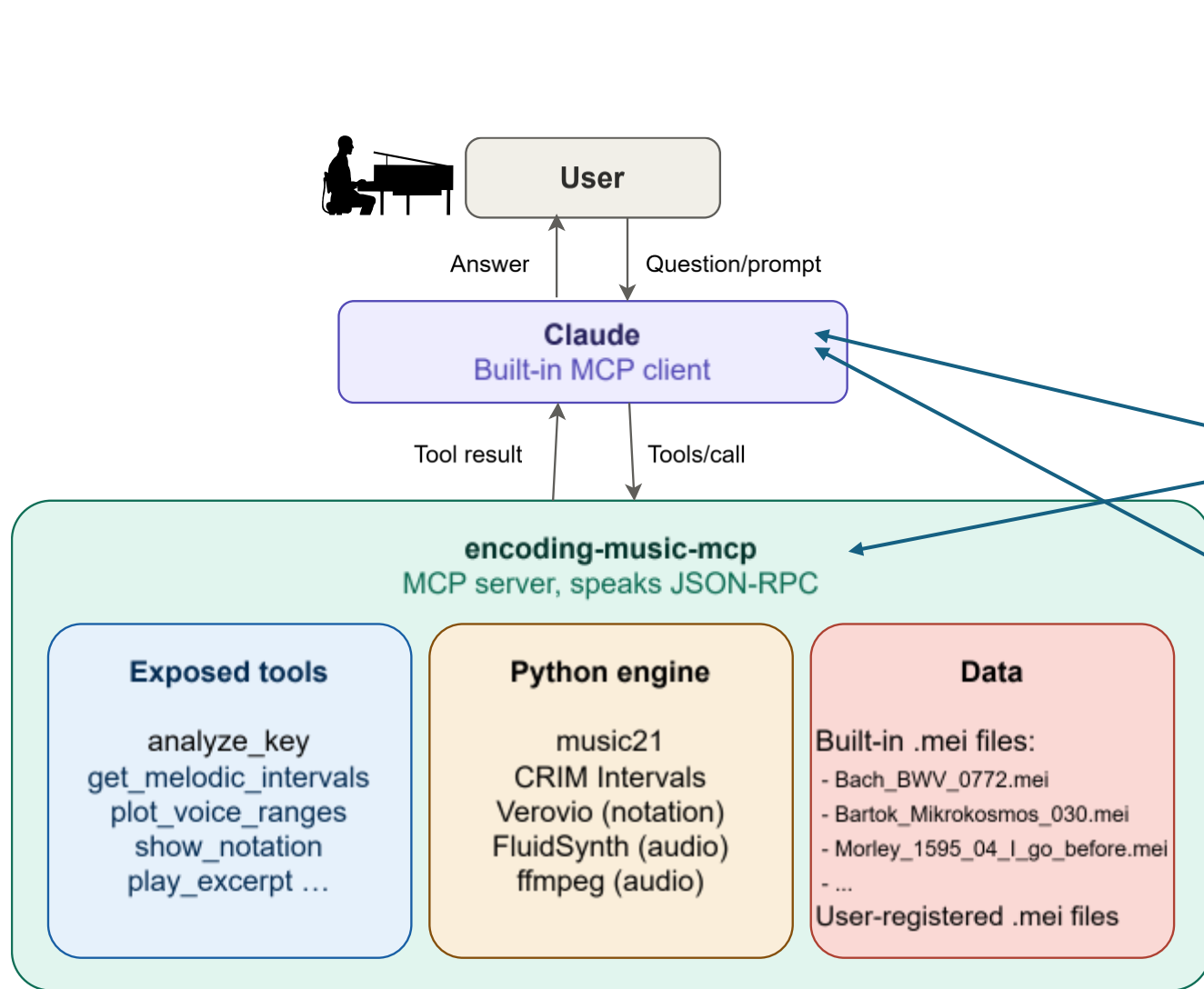
Physical and intellectual characteristics
of music notation documents

eXtensible Markup Language (XML) schema

The image displays a musical score for the song "Im wunderschönen Monat Mai" by Franz Schubert. The score is written for voice and piano. The lyrics are: "Im wunderschönen Monat Mai, als alle Knospen der schönen Tage sich zu öffnen beginnen." The XML schema is overlaid on the right side of the score, showing the encoding of the musical notation. The schema includes elements for measures, staves, layers, notes, verses, and syllables, with attributes for pitch, duration, and other musical characteristics.

```
<measure n="5">
  <staff n="1">
    <layer n="1">
      <note pname="c" oct="5" dur="8" dots="1" stem.dir="down" accid.ges="s">
        <verse n="1">
          <syl wordpos="i" con="d">Wun</syl>
        </verse>
      </note>
      <note pname="b" oct="4" dur="16" stem.dir="down">
        <verse n="1">
          <syl wordpos="m" con="d">der</syl>
        </verse>
      </note>
      <note pname="b" oct="4" dur="16" stem.dir="up">
        <verse n="1">
          <syl wordpos="m" con="d">schö</syl>
        </verse>
      </note>
      <note pname="d" oct="5" dur="16" stem.dir="up">
        <verse n="1">
          <syl wordpos="t">nen</syl>
        </verse>
      </note>
      <note pname="f" oct="4" dur="16" stem.dir="up" accid.ges="s">
```



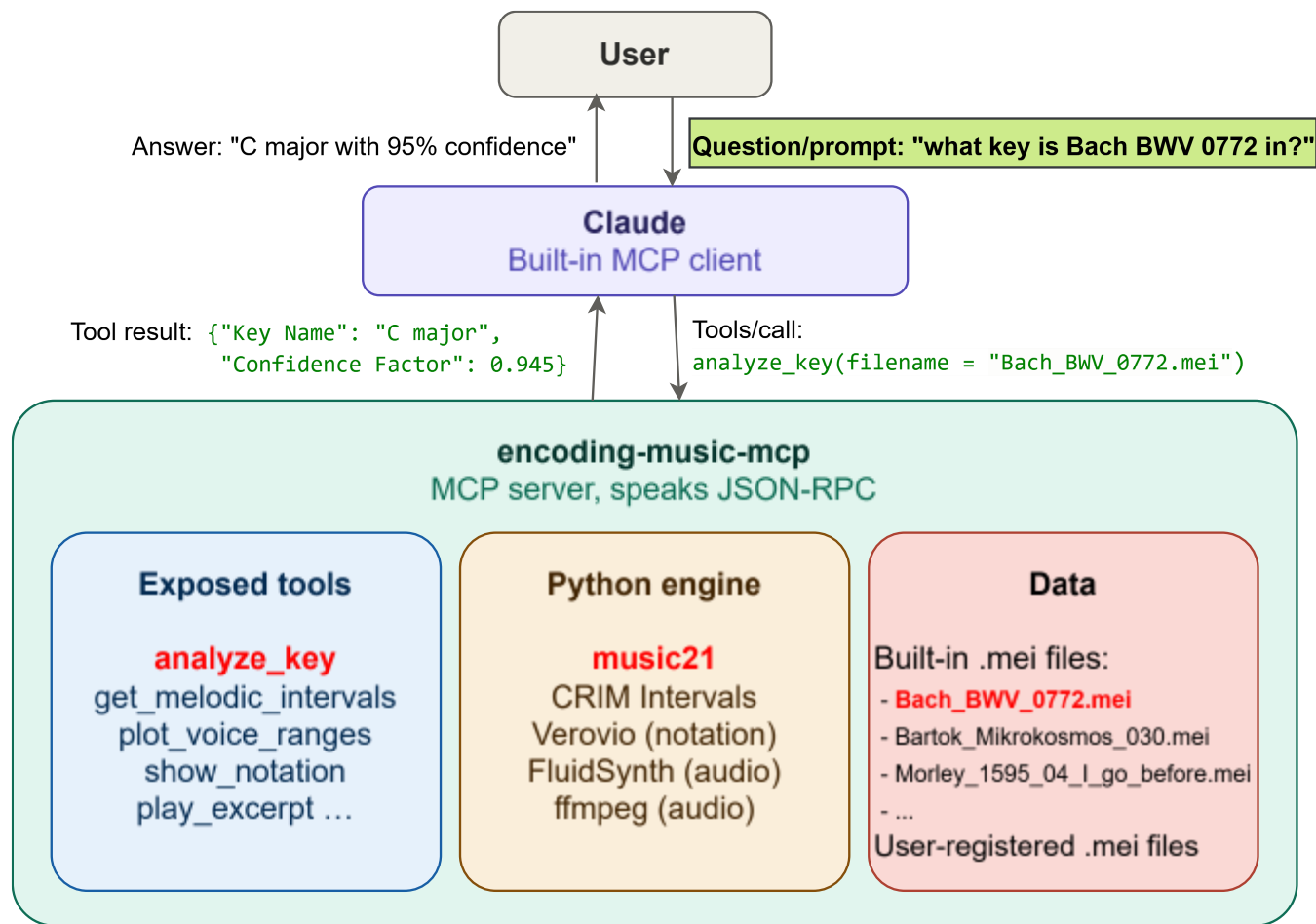


Model Context Protocol



Claude

User asks questions in natural language



Bach BWV 772 key identification ▾

what key is Bach BWV 0772 in?

Loaded tools, used encoding-music-mcp integration ▾

🕒 I know Bach's Invention No. 1 in C major is BWV 772, but let me verify this using the music analysis tool I have access to since there's an MEI file available for the piece.

🔍 Loading tools

Result

E Analyze key

Result

✅ Done

Bach's Invention No. 1, BWV 772, is in **C major** — analysis of the score gives this with very high confidence (~0.95).

📄 👍 💬 ↺

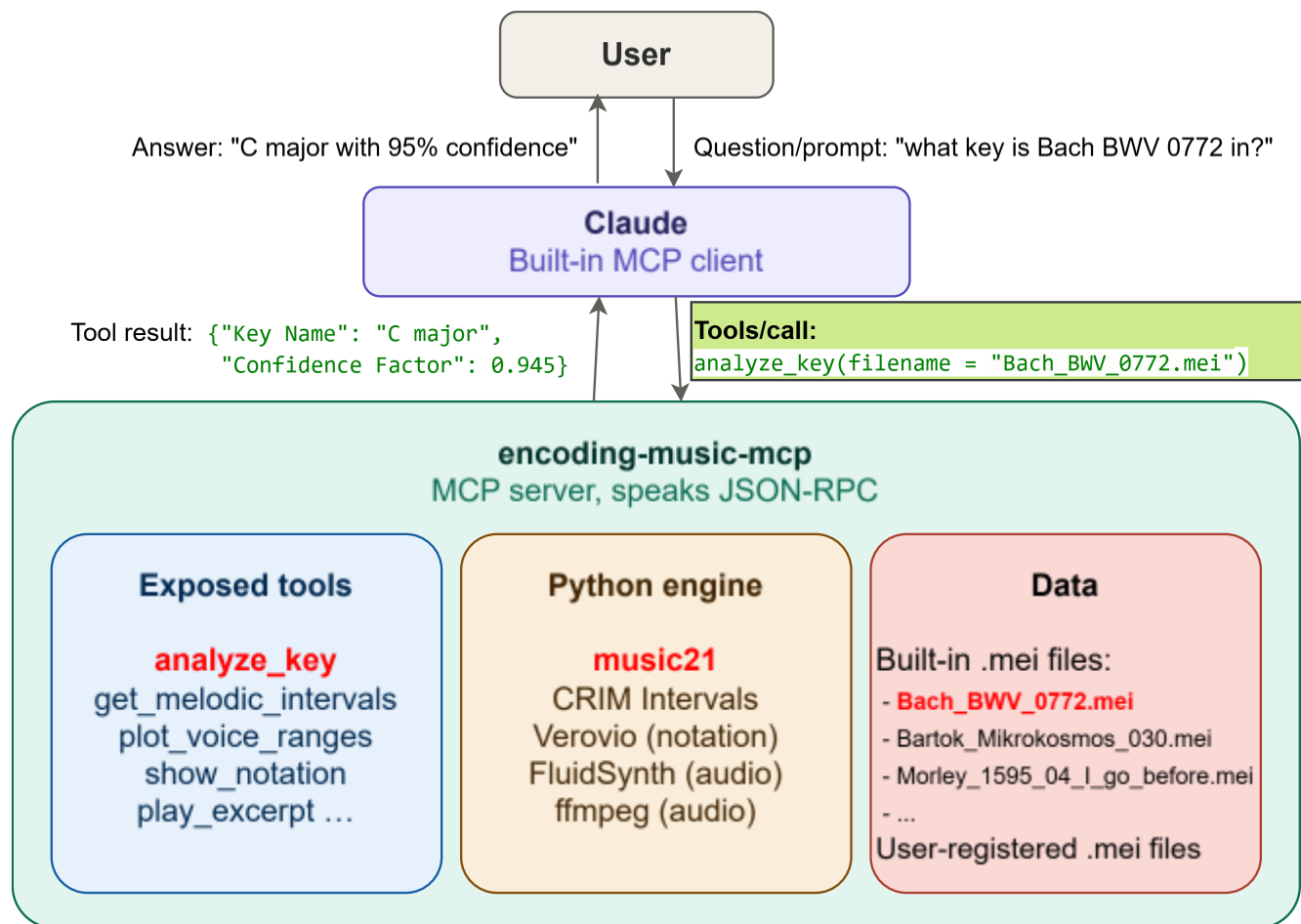
✳️

Write a message...

+ Opus 4.7 Adaptive ▾ 🎤

Claude is AI and can make mistakes. Please double-check responses.

MCP client decides which tool(s) to call



Bach BWV 772 key identification ▾

what key is Bach BWV 0772 in?

Loaded tools, used encoding-music-mcp integration ▾

I know Bach's Invention No. 1 in C major is BWV 772, but let me verify this using the music analysis tool I have access to since there's an MEI file available for the piece.

Loading tools

Result

Analyze key

Request

```
{
  "filename": "Bach_BWV_0772.mei"
}
```

Response

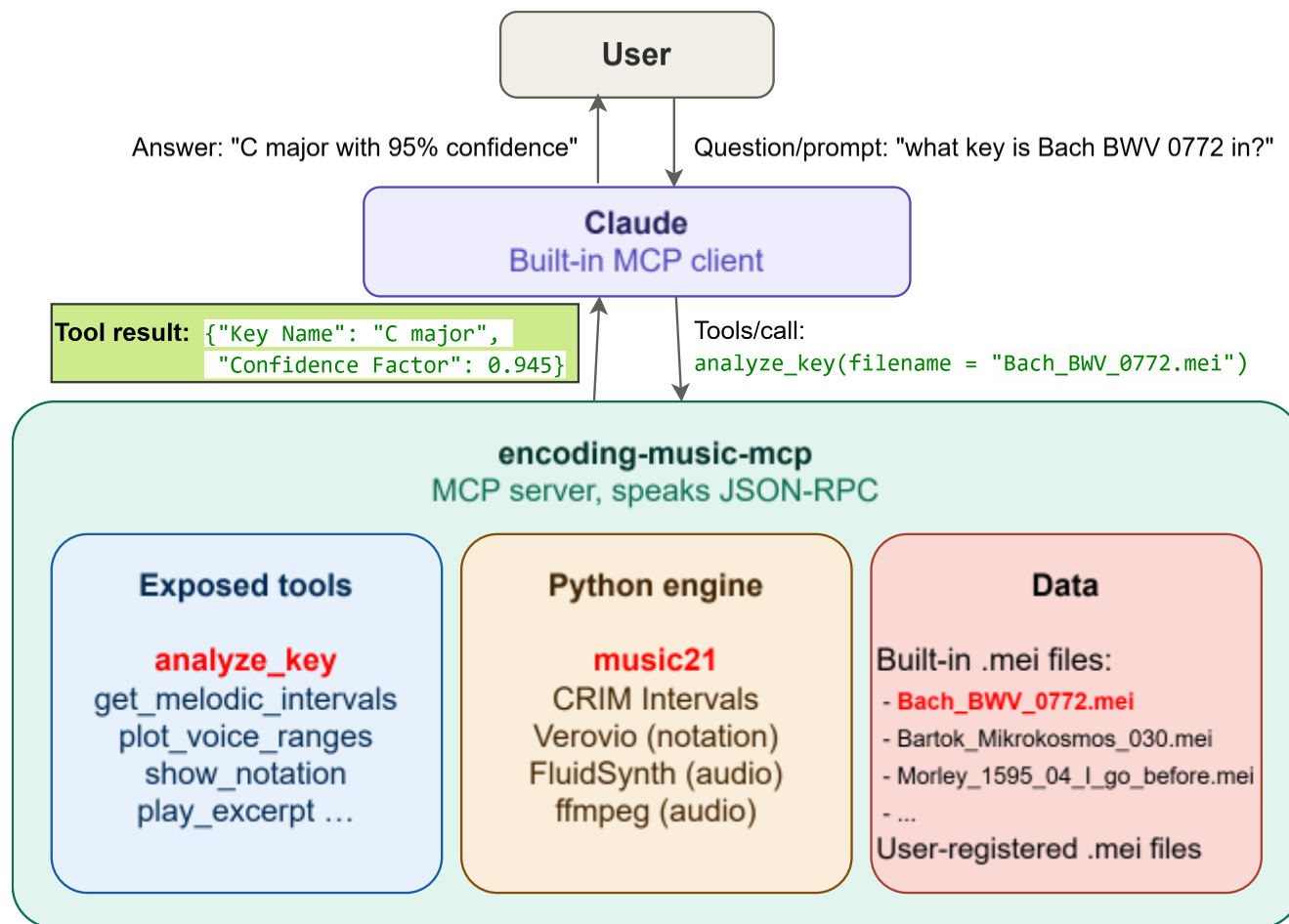
```
{"Key Name": "C major", "Confidence Factor": 0.9450617236680577}
```

Write a message...

Opus 4.7 Adaptive ▾

Claude is AI and can make mistakes. Please double-check responses.

MCP server returns the tool result



Bach BWV 772 key identification ▾

what key is Bach BWV 0772 in?

Loaded tools, used encoding-music-mcp integration ▾

🕒 I know Bach's Invention No. 1 in C major is BWV 772, but let me verify this using the music analysis tool I have access to since there's an MEI file available for the piece.

🔍 Loading tools

Result

E Analyze key

Request

```
{
  "filename": "Bach_BWV_0772.mei"
}
```

Response

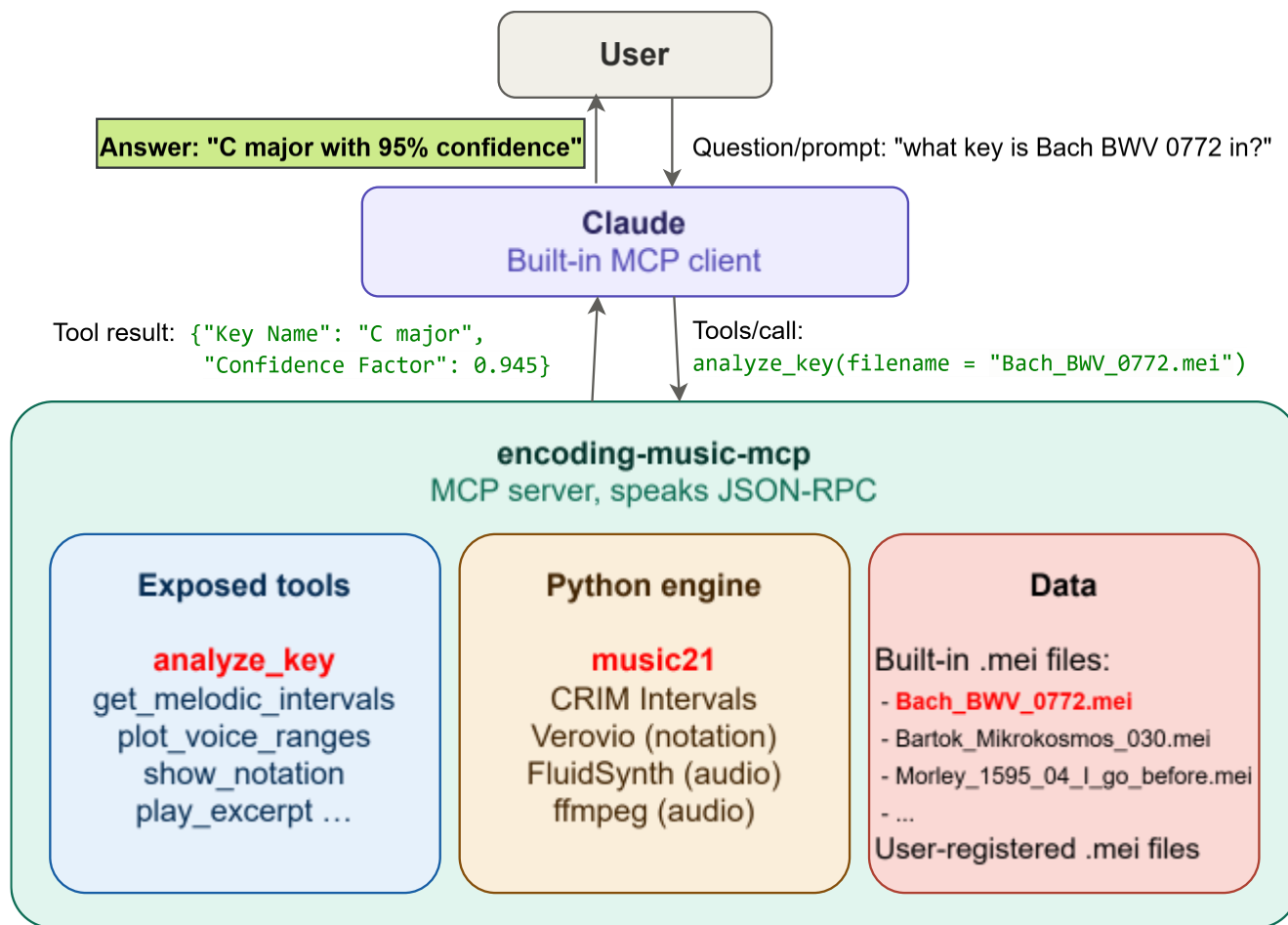
```
{"Key Name": "C major", "Confidence Factor": 0.9450617236680577}
```

Write a message...

+ Opus 4.7 Adaptive ▾ 🎤

Claude is AI and can make mistakes. Please double-check responses.

MCP client interprets the result



Bach BWV 772 key identification ▾

what key is Bach BWV 0772 in?

Loaded tools, used encoding-music-mcp integration >

Bach's BWV 772 (Invention No. 1) is in **C major**, with a confidence factor of 0.95 from the music21 key analysis.

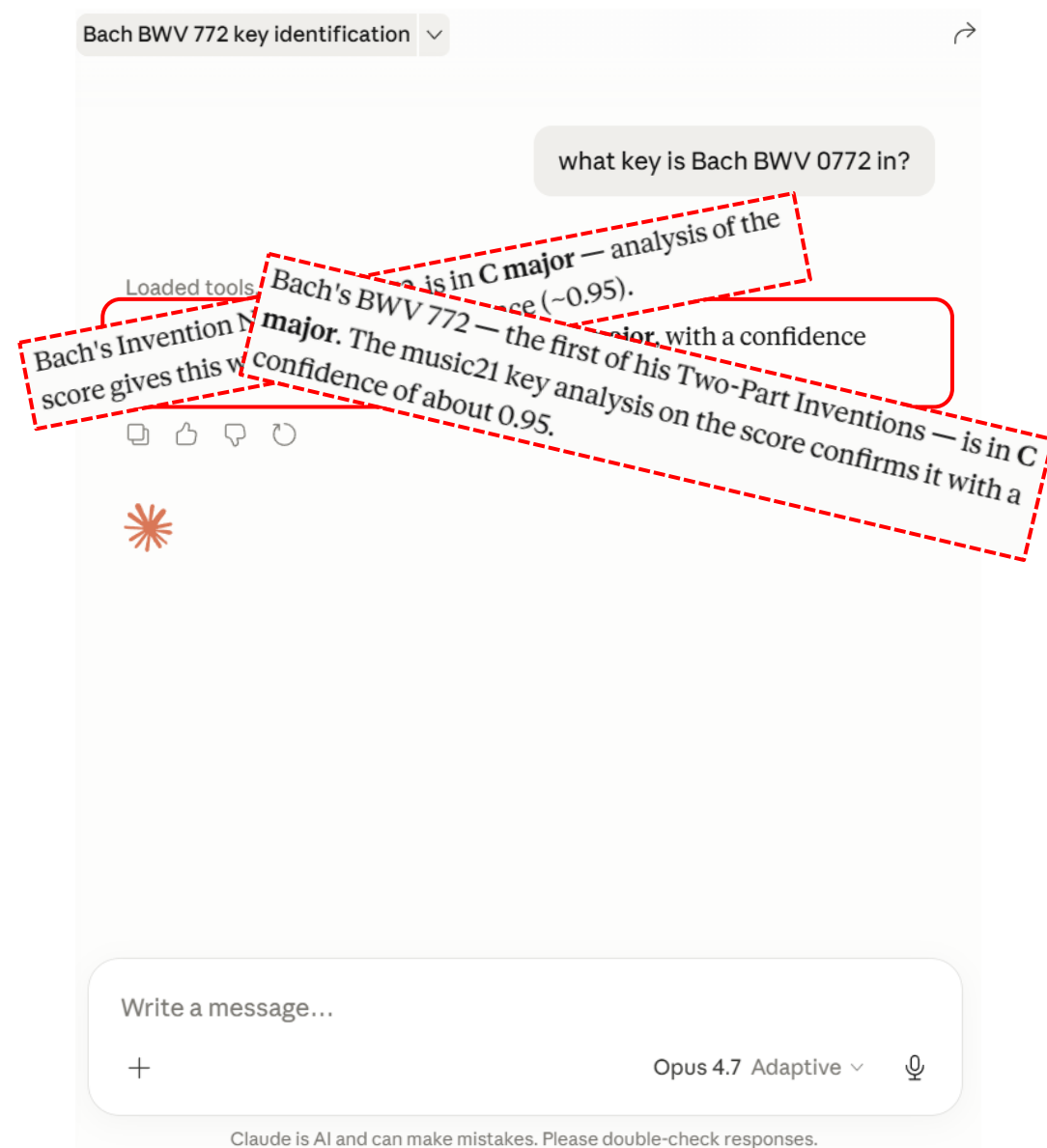
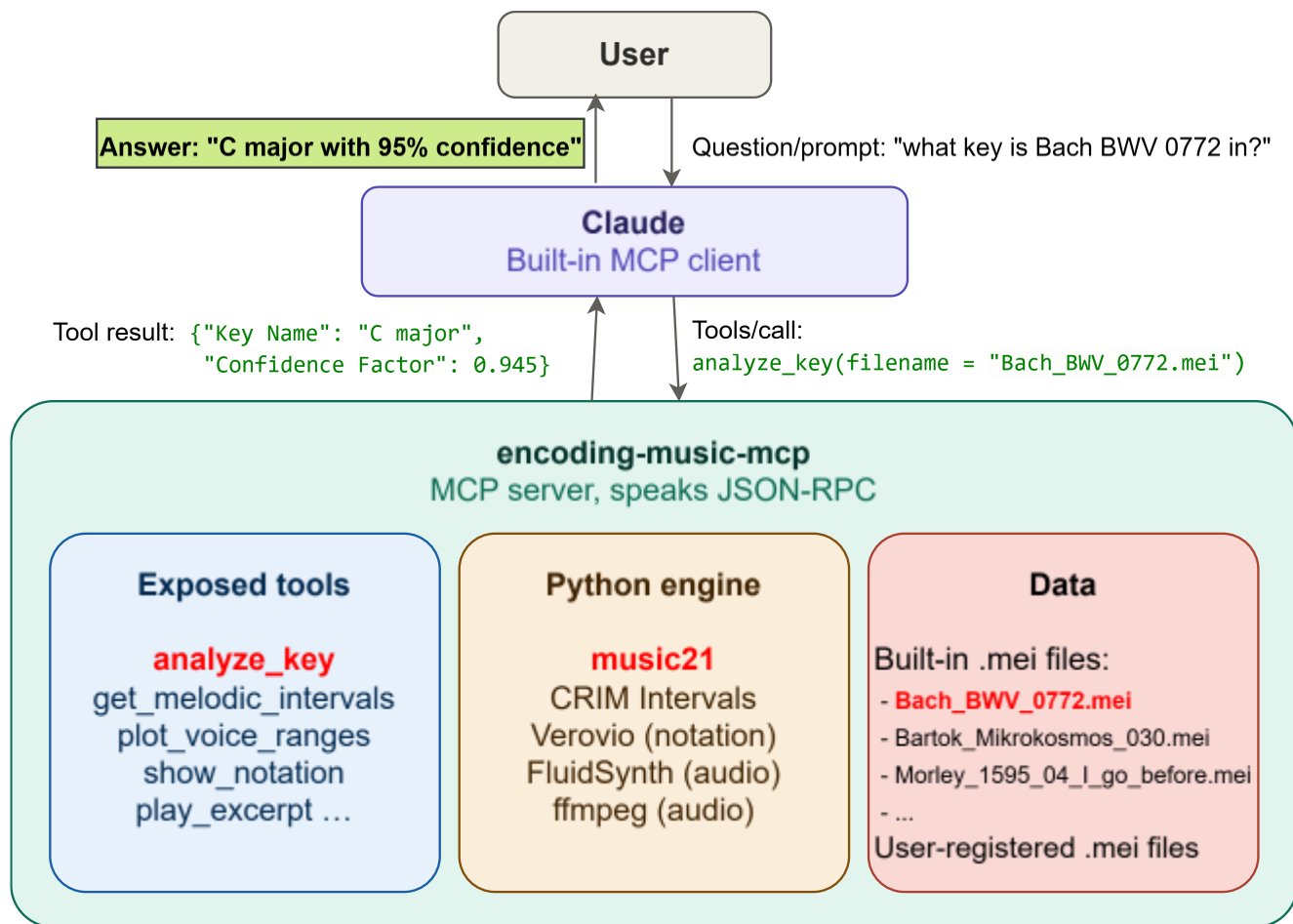
✱

Write a message...

Opus 4.7 Adaptive ▾

Claude is AI and can make mistakes. Please double-check responses.

MCP client interprets the result



Visualisation tools

📄 Melodic n-gram heatmap

Used encoding-music-mcp integration, loaded tools >

E encoding-music-mcp plot_melodic_ngram_heatmap

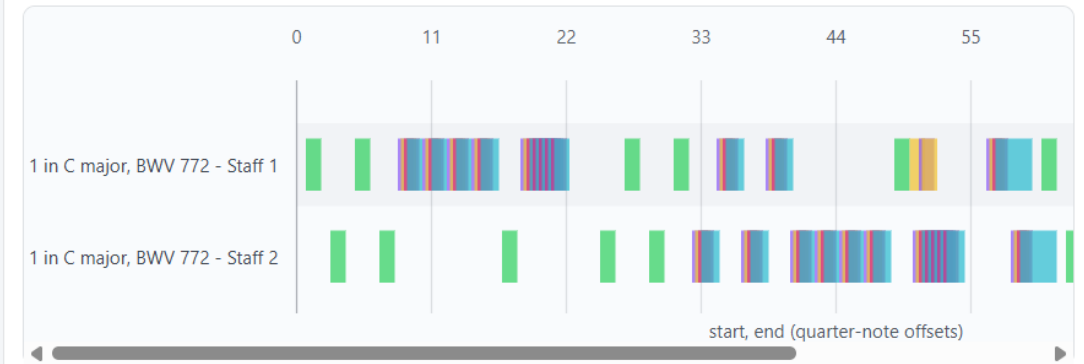
</>

Melodic N-gram Heatmap

1 score

- Bach, Johann Sebastian | Invention No. 1 in C major, BWV 772

n=4 | kind=d | top 5 patterns across all selected scores.



■ **-2_3_-2_3**

27 occurrences

■ **3_-2_3_-2**

27 occurrences

■ **-2_-2_3_-2**

21 occurrences

■ **-2_-2_-2_3**

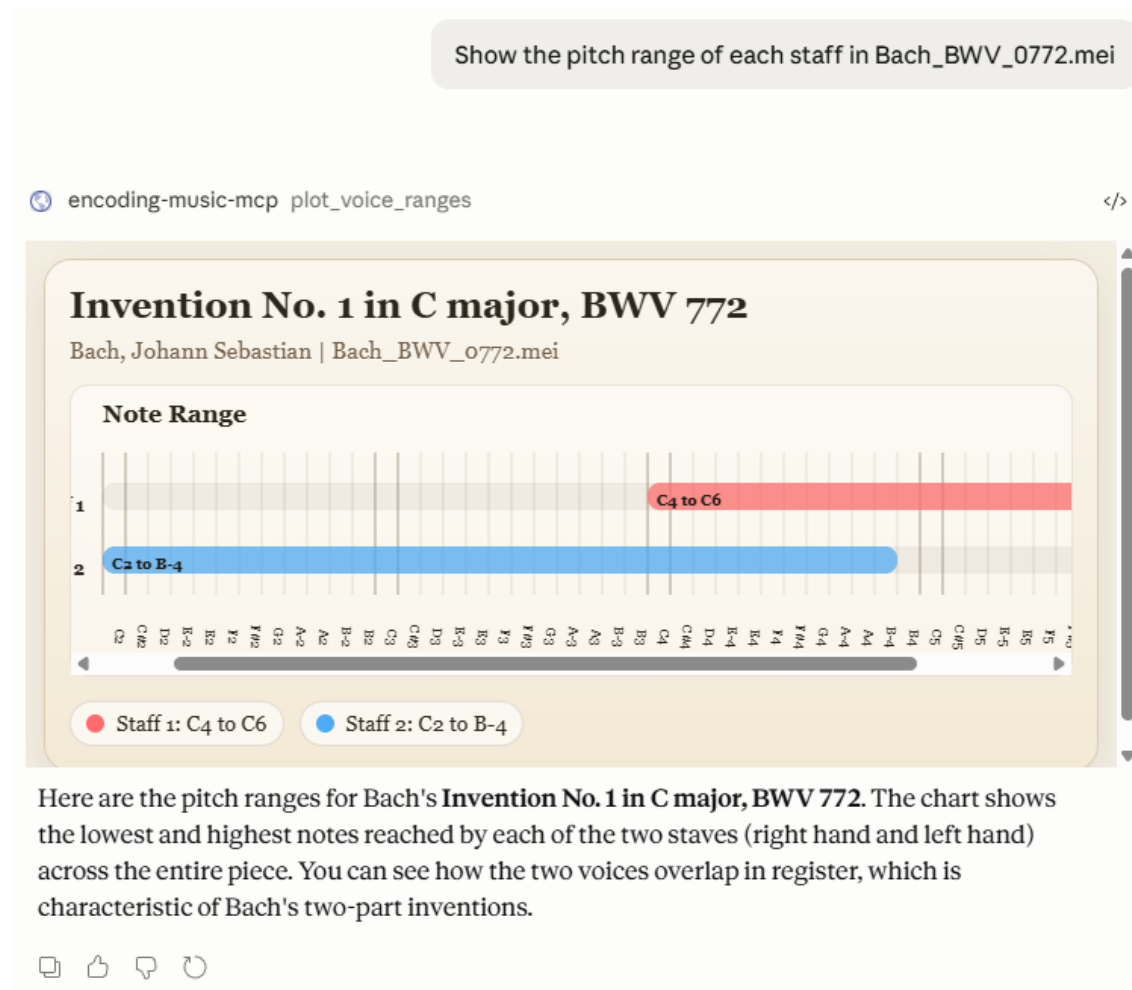
20 occurrences

■ **2_-3_2_-3**

20 occurrences

Visualisation tools

- ☐ Melodic n-gram heatmap
- ☐ Pitch range



Visualisation tools

- ☐ Melodic n-gram heatmap
- ☐ Pitch range
- ☐ Weighted note distribution

weighted note distributions for Bach_BWV_0772.mei and Bach_BWV_0773.mei

encoding-music-mcp plot_weighted_note_distribution

</>

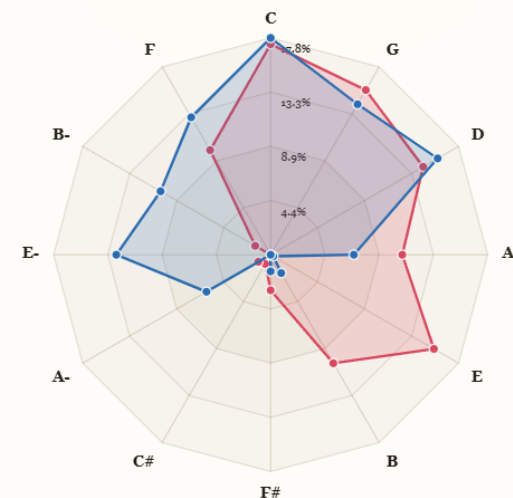
2 scores

Bach, Johann Sebastian | Invention No. 1 in C major, BWV 772

Bach, Johann Sebastian | Invention No. 2 in C minor, BWV 773

Scaled by note duration, ordered by fifths, one overlaid polygon per score.

Weighted Note Distribution



Traces

● Invention No. 1 in C major, BWV 772

466 sounded note events

C 17.2% | G 15.6% | E 15.4%

The radar chart overlays both inventions for easy comparison:

- **BWV 772 (C major)** is centred on C and G, with a fairly symmetrical profile around the tonic.
- **BWV 773 (C minor)** shifts the weight toward E ♭, B ♭, and G, reflecting its minor key — the flattened third and seventh are much more prominent.

The contrast between the two profiles nicely illustrates how the tonal centre and mode shape the overall pitch-class usage across an entire piece.

Visualisation tools

- ☐ Melodic n-gram heatmap
- ☐ Pitch range
- ☐ Weighted note distribution
- ☐ Sonority n-gram progress



Here's the sonority n-gram progress plot for **CRIM_Model_0012.mei** — Pierre Sandrin's *Voulant honneur*.

A few highlights from the analysis:

- **104 distinct sonority n-gram patterns** were found across **258 total occurrences**, plotted from 0% to 99% of the score's timeline.
- The most recurring patterns (4 occurrences each) are dense, sustained chord sequences — notice the many `HeId_12/8/...` entries, reflecting passages where voices sustain notes across beats, producing stable vertical sonorities.
- The pattern `-M2_10/5/1_-P5_12/8/3/1_...` appears first right at the opening (~0%), suggesting a characteristic opening gesture that recurs throughout.
- The distribution of patterns across the score timeline gives a sense of how Sandrin recycles and varies his harmonic language from beginning to end.

Visualisation tools

- ☐ Melodic n-gram heatmap
- ☐ Pitch range
- ☐ Weighted note distribution
- ☐ Sonority n-gram progress

Notation tools

- ☐ Show notation

Register C:\Users\jaman\Downloads\CRIM_Model_0002.mei and show notation

It was already registered in our previous conversation, but I'll re-register it and show the notation now.

Register mei file from path >

 encoding-music-mcp show_notation

</>

O gente brunette



Superius

Contratenor

Tenor

Bassus

← Prev

Page 1 of 11

Next →

CRIM_Model_0002.mei is registered and displayed — 11 pages in total. Use the widget to navigate through the score. Let me know if you'd like any analysis!

Visualisation tools

- ☐ Melodic n-gram heatmap
- ☐ Pitch range
- ☐ Weighted note distribution
- ☐ Sonority n-gram progress

Notation tools

- ☐ Show notation
- ☒ Show notation with highlight

Highlight the top 2 melodic 4-grams in Bach BWV 0772 in full notation

Synthesizing overlapping melodic patterns and deduplicating note IDs >

The top 2 are tied at 27 occurrences each: **-2_3_-2_3** and **3_-2_3_-2** — both rotations of the iconic zigzag motive in Invention No. 1. Let me get all occurrences.

Consolidated overlapping melodic patterns and deduplicated note identifiers >

Now to highlight all unique notes from both patterns. Let me dedupe them with a quick script.

Dedupe note IDs from both top-2 melodic 4-gram patterns >

🌐 encoding-music-mcp show_notation_highlight

</p>

Highlighted 11 notes

Invention No. 1 in C major, BWV 772



← Prev

Page 1 of 7

Next →

Both top patterns are highlighted — 130 unique notes across 54 occurrences (27 each of $-2\ 3\ -2\ 3$ and $3\ -2\ 3\ -2$).

These are the two rotations of the famous opening zigzag from Invention No. 1: step-down-then-skip-up vs. skip-up-then-step-down. Because they're offset by just one note, almost every appearance of the figure lights up under both patterns simultaneously, which is why the highlighting blankets the running sixteenth-note lines for long stretches. Page through the widget and you'll see the motive everywhere — it's the rhythmic and melodic DNA of the whole piece. The unhighlighted moments are where Bach breaks the figure: cadential gestures, the secondary motive ($\underline{2_3_2_3} / \underline{-3_2_3_2}$, the next-ranked patterns), and the suspension-resolution figures around structural arrivals.

Visualisation tools

- ☐ Melodic n-gram heatmap
- ☐ Pitch range
- ☐ Weighted note distribution
- ☐ Sonority n-gram progress

Notation tools

- ☐ Show notation
- ☐ Show notation with highlight
- ☐ Show notation with elicitation

Your plan ends in 23 days · [Resubscribe](#)

 Afternoon, Anqi

How can I help you today?

+

Sonnet 4.6 ▾ 

 Code

 Learn

 Life stuff

 Write

 Claude's choice

Visualisation tools

- ☐ Melodic n-gram heatmap
- ☐ Pitch range
- ☐ Weighted note distribution
- ☐ Sonority n-gram progress

Notation tools

- ☐ Show notation
- ☐ Show notation with highlight
- ☐ Show notation with elicitation

Playback tools

- ☐ Playback
- ☐ Playback with elicitation

Play the score Bach BWV 772 with a bpm of 70

Determining filename format for Bach score playback >

 encoding-music-mcp play_excerpt </>

Bach_BWV_0772.mei | full piece | 70 bpm | 78.78s

▶ 1:18 / 1:18

🔊 ⋮

Audio loaded

Here's Bach's Invention No. 1 in C major (BWV 772) at 70 bpm. Let me know if you'd like a different tempo, just a specific passage, or to see the notation alongside it.



Future works

- ☐ More visualisation tools
- ☐ Presentation type tool
- ☐ A tool to validate user-registered .mei files
- ☐ A function allowing users to specify a bar range for focused analysis
- ☐ ...

encoding-music-mcp [<https://github.com/unimelbmdap/encoding-music-mcp>]

Reflection and Key Learnings

- ☐ Music score analysis
- ☐ Working with GitHub
- ☐ Test-driven development
- ☐ Designing tools for non-technical users
- ☐ Data visualisation and interface design

Thank you MDAP!

Questions?